Red ASM Security

SOFTWARE DESIGN DOCUMENT (SDD)

SP-108-RedASM

CS 4850 – Computer Science Senior Project

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1. Introduction

1.1 Document Outline

This Software Design Document (SDD) provides a detailed description of the architecture, design decisions, components, and operational constraints of the Red ASM Security Cyber Attack Surface Management (ASM) project.

1.2 Document Description

1.2.1 Introduction

Red ASM Security is a hands-on cyber security project designed to demonstrate how attack surfaces evolve over time due to software drift, misconfiguration, and shadow IT. The system simulates an enterprise environment and a Security Operations Center (SOC) using cloud-based virtual machines and open-source security tooling.

The primary goal is educational: to show how automated discovery, monitoring, alerting, and response mechanisms can be used to identify and mitigate security risks introduced by unmanaged or poorly managed assets.

1.2.2 System Overview

The system consists of two virtual machines deployed in Oracle Cloud Infrastructure (OCI):
- **Target VM** – Represents an enterprise system with intentional and unintentional exposure.
- **Manager VM** – Represents a SOC platform providing SIEM and SOAR capabilities.

The Manager VM centrally monitors the Target VM using Wazuh (SIEM) and Shuffle (SOAR), enabling detection, alerting, and automated response workflows.

2. Design Considerations

2.1 Assumptions and Dependencies

- Oracle Cloud Infrastructure is available and provides free-tier resources.
- Ubuntu 22.04 Minimal (ARM64) is used as the operating system.
- Users have basic familiarity with Linux system administration.
- Internet connectivity is available for dashboards, updates, and alerting.
- External services such as Slack and Microsoft Azure are reachable.
- Wazuh, Shuffle, Docker, and Microsoft Graph APIs remain stable.

2.2 General Constraints

- Limited hardware resources (2 CPUs, 12 GB RAM per VM).
- Manager VM is isolated on a private subnet.
- Network access restricted to team IP addresses.
- Use of lightweight desktop environments (XFCE).
- Compatibility constraints between desktop environments and RDP services.
- Security policies restrict SSH and RDP ports.

2.3 Goals and Guidelines

- Demonstrate real-world ASM concepts.
- Maintain minimal system overhead.
- Prioritize security and access control.
- Favor open-source and industry-relevant tools.
- Ensure reproducibility of the environment.

2.4 Development Methods

The project follows an **iterative and experimental development approach**, combining: - Incremental system configuration - Trial-and-error validation - Documentation-driven refinement

This approach was chosen due to the complexity of integrating multiple security tools and cloud constraints.

3. Architectural Strategies

- **Two-tier architecture** separating monitored assets and SOC tooling.
- **Agent–Manager model** using Wazuh.
- **Container-based orchestration** for Shuffle using Docker.
- **Zero-trust networking principles** via subnet isolation.
- **Least-privilege access** using Linux users, groups, and SSH keys.
- **Automation-first response design** using SOAR workflows.

Alternative architectures (single VM, GUI-heavy OS, Ubuntu 24.04) were rejected due to performance issues, desktop conflicts, and reduced system stability.

4. System Architecture

The RedASM system is decomposed into two major subsystems:

1. **ASM Target Subsystem**
2. **SOC Manager Subsystem**

These subsystems communicate over a private virtual network and are logically isolated for security.

4.1 Subsystem Architecture

4.1.1 ASM Target VM

- Public-facing VM
- Hosts intentionally exposed services
- Runs Wazuh Agent
- Acts as a bastion host

4.1.2 Manager VM

- Private subnet only
- Hosts Wazuh Manager, Indexer, and Dashboard
- Hosts Shuffle SOAR platform
- Handles alerting, automation, and response

5. Policies and Tactics

- Network security via iptables and OCI security lists
- SSH key-based authentication only
- Role-based access using Linux groups
- Automated monitoring and alerting
- Centralized logging and event correlation

6. Detailed System Design

6.1 ASM Target Subsystem

Classification

Subsystem / Virtual Machine

Definition

Simulates an enterprise endpoint with evolving attack surface exposure.

Responsibilities

- Host monitored directories
- Expose services
- Send logs and alerts to Manager VM

Constraints

- Limited CPU and RAM
- Public IP exposure
- Restricted inbound access

Composition

- Ubuntu 22.04 Minimal
- XFCE4 Desktop
- XRDP + Xorg
- Wazuh Agent

Uses / Interactions

- Communicates with Wazuh Manager
- Accessible via SSH and RDP

Resources

- CPU, RAM, disk storage
- Network ports 22 and 3389

Processing

- File integrity monitoring
- Log forwarding

Interface / Exports

- Wazuh Agent events
- SSH and RDP access

6.2 SOC Manager Subsystem

Classification

Subsystem / Virtual Machine

Definition

Centralized SOC environment providing SIEM and SOAR capabilities.

Responsibilities

- Event correlation
- Alert generation
- Automated response workflows
- Notification delivery

Constraints

- Private network only
- Port conflicts between services
- External authentication dependencies

Composition

- Wazuh Manager
- Wazuh Indexer
- Shuffle (Docker-based)
- Slack and Outlook integrations

Uses / Interactions

- Receives data from Target VM
- Sends alerts to Slack and Exchange email

Resources

- Docker containers
- External APIs (Microsoft Graph)
- Authentication tokens

Processing

- Rule-based alerting
- Workflow execution
- Token refresh logic

Interface / Exports

- Web dashboards
- REST-based workflow triggers
- Email and Slack notifications

6.3 Shared Project Resources

Classification

Filesystem / Security Component

Definition

Shared project workspace for team collaboration.

Responsibilities

- Secure storage of project artifacts
- Controlled access

Constraints

- Group-based permissions

Composition

- /srv/asm_project
- asm_project Linux group

7. Glossary

- **ASM** – Attack Surface Management
- **SIEM** – Security Information and Event Management
- **SOAR** – Security Orchestration, Automation, and Response
- **Shadow IT** – Unmanaged or unauthorized IT assets
- **Bastion Host** – Gateway system for secure access
- **Wazuh** – Open-source SIEM platform
- **Shuffle** – Open-source SOAR platform